

Section 7.1

Exercise (7.1.7). The center of a ring R is $\{z \in R \mid zr = rz \text{ for all } r \in R\}$. Prove that the center of a ring is a subgroup that contains the identity. Prove that the center of a division ring is a field.

Proof: Suppose a ring R with identity and let $G = \{z \in R \mid zr = rz \text{ for all } r \in R\}$. Note that by definition we get that $1 \in G$ since $1r = r1$ for all $r \in R$. Let $a, b^{-1} \in G$ and $r \in R$, note that $(a + b^{-1})r = (ar + b^{-1}r) = (ra + rb^{-1}) = r(a + b^{-1})$. Therefore $G \leq R$ with respect to addition. Note that for all $r \in R$ we have that $(ab)r = (ar)b = r(ab)$. Therefore G is closed with respect to multiplication and thus is a subring of R .

Suppose R is a division ring and let $g \in G$ and note that $g^{-1}r = (r^{-1}g)^{-1} = (gr^{-1})^{-1} = rg^{-1}$ so $g^{-1} \in G$. Therefore G is a commutative division ring, and therefore a field. $\square$

Exercise (7.1.9). For a fixed element $a \in R$ define $C(a) = \{r \in R \mid ra = ar\}$. Prove that $C(a)$ is a subgroup of R containing a . Prove that the center of R is the intersection of the subrings $C(a)$ over all $a \in R$.

Proof: Suppose $a \in R$ and define $C(a) = \{r \in R \mid ra = ar\}$. Clearly $a \in C(a)$ since $a(a) = (a)a$. Let $a, b^{-1} \in C(a)$ and note that $(a + b^{-1})r = (ar + b^{-1}r) = (ra + rb^{-1}) = r(a + b^{-1})$ so $C(a) \leq R$ with respect to addition.

Let $G = \bigcap_{a \in R} C(a)$, and $Z = \{z \in R \mid zr = rz \text{ for all } r \in R\}$. Suppose $g \in G$ and let $a \in R$, by definition $g \in C(a)$ so $ga = ag$. Therefore $g \in Z$ so $G \subseteq Z$. Suppose $z \in Z$, by definition for all $a \in R$ we know that $za = az$ therefore $z \in C(a)$ for all $a \in R$. Thus $z \in G$ and therefore $G = Z$. $\square$

Exercise (7.1.11). Prove that if R is an integral domain and $x^2 = 1$ for some $x \in R$ then $x = \pm 1$.

Proof: Suppose R is an integral domain with $x^2 = 1$ for some $x \in R$. Note that $x^2 - 1 = 0$, so therefore $(x - 1)(x + 1) = x^2 - 1 = 0$. By definition R has no zero divisors, therefore $x - 1 = 0$ or $x + 1 = 0$ so $x = \pm 1$. $\square$

Exercise (7.1.13). An element x in R is called nilpotent if $x^m = 0$ for some $m \in \mathbb{Z}^+$.

- (a) Show that if $n = a^k b$ for some integers a and b then $\overline{ab}$ is a nilpotent element of $\mathbb{Z}/n\mathbb{Z}$.

Proof: Suppose $n = a^k b$ for some integers a and b and consider the ring $\mathbb{Z}/n\mathbb{Z}$. Note the following $\overline{ab}$,

$$\overline{ab}^k = \overline{a^k b^k} = (\overline{a^k} \overline{b^k}) \overline{b}^{k-1} = (0) \overline{b}^{k-1} = 0.$$

Thus $\overline{ab}$ is nilpotent in $\mathbb{Z}/n\mathbb{Z}$. □

- (b) If $a \in \mathbb{Z}$ is an integer, show that the element $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is nilpotent if and only if every prime divisor of n is also a divisor of a . In particular, determine the nilpotent elements of $\mathbb{Z}/72\mathbb{Z}$ explicitly.

Proof: Suppose $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is nilpotent. Then there exists some $m \in \mathbb{Z}^+$ such that $\bar{a}^m = 0$. Therefore we know that $n \mid a^m$ so every prime divisor of n must also divide a^m and therefore a .

Suppose that $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ and every prime divisor of n is also a divisor of a . Therefore $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_i^{\alpha_i}$ has some prime factorization, now choose m such that $a^m = p_1^{\beta_1} p_2^{\beta_2} \dots p_i^{\beta_i} (l)$ has the property that $\beta_i \geq \alpha_i$ and l is some integer with no common prime divisors. Finally note that,

$$a^m = p_1^{\beta_1} p_2^{\beta_2} \dots p_i^{\beta_i} (l) = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_i^{\alpha_i} (p_1^{\beta_1 - \alpha_1} p_2^{\beta_2 - \alpha_2} \dots p_i^{\beta_i - \alpha_i} (l)) = n (p_1^{\beta_1 - \alpha_1} p_2^{\beta_2 - \alpha_2} \dots p_i^{\beta_i - \alpha_i} (l)).$$

Thus $\bar{a}^m = 0$ and therefore $\bar{a}$ is nilpotent in $\mathbb{Z}/n\mathbb{Z}$.

By the previous result every nilpotent element in $\mathbb{Z}/72\mathbb{Z}$ must have the same prime divisors as $72 = 2^3 3^2$. So every nilpotent element must factor 6, which gives us the following elements, $\bar{6}, \bar{12}, \bar{18}, \bar{24}, \bar{30}, \bar{36}, \bar{42}, \bar{48}, \bar{54}, \bar{60}, \bar{66}$. □

- (c) Let R be the ring of functions from a nonempty set X to a field F . Prove that R contains no nonzero nilpotent elements.

Proof: Suppose R be the ring of functions from a nonempty set X to a field F and for the sake of contradiction let $f \in R$ such that $f \neq 0$ and f is nilpotent. Since f nilpotent and a function from a nonempty set X to a field F there exists some m such that for all $x \in X$, $f(x)^m = 0$ i.e, f^m is the zero map in R . Consider $a \in X$ such that $f(a) \neq 0$ and $f(a)^{m-1} \neq 0$. and note that $f(a)f(a)^{m-1} = f(a)^m = 0$. Therefore there exists zero divisors in F a contradiction. $\square$

Exercise (7.1.14). Let x be a nilpotent element of the commutative ring R .

- (a) Prove that x is either zero or a zero divisor.

Proof: Suppose $x \in R$ is nilpotent and R is commutative. Let m be minimal where $x^m = 0$ then it follows that $xx^{m-1} = 0$. Suppose $x \neq 0$, since m is minimal $x^{m-1} \neq 0$ and therefore x is a zero divisor. $\square$

- (b) Prove that rx is nilpotent for all $r \in R$.

Proof: Suppose $x \in R$ is nilpotent and R is commutative. Let m such that $x^m = 0$. Note that $(rx)^m = r^m x^m = r^m(0) = 0$ thus rx is nilpotent. $\square$

- (c) Prove that $1 + x$ is a unit in R .

Proof: Suppose $x \in R$ is nilpotent and R is commutative. $(1+x)(1+\sum_{i=1}^{m-1}(-1)^i x^i) = (1)(1+\sum_{i=1}^{m-1}(-1)^i x^i) + (x)(1+\sum_{i=1}^{m-1}(-1)^i x^i) = 1 + \sum_{i=1}^{m-1}(-1)^i x^i + \sum_{i=1}^m(-1)^{i+1} x^i = 1 \pm x^m = 1$. $\square$

- (d) Deduce that the sum of a nilpotent element and a unit is a unit.

Proof: Suppose $x \in R$ is nilpotent, $r \in R$ is unit and R is commutative. There exists an $a \in R$ such that $ra = ar = 1$, and some m such that $x^m = 0$. By (b) we know that ax is also nilpotent. By (c) we know that $(1 + ax)$ is a unit such that for $b \in R$, $(1 + ax)b = b(1 + ax) = 1$. Finally note that $(x + u) = (xa + 1)u$. Multiplying by (ab) , since R is commutative we get, $(x + u)(ab) = (xa + 1)u(ab) = ((xa + 1)b)(ua) = 1$ $\square$

Exercise (7.1.28). Let R be a ring with $1 \neq 0$. a nonzero element a is called left zero divisor in R if there is a nonzero element $x \in R$ such that $ax = 0$. Symmetrically, $b \neq 0$ is a right zero divisor if there is a nonzero $y \in R$ such that $yb = 0$. An element $u \in R$ has left inverse in R if there is some $s \in R$ such that $su = 1$. symmetrically v has a right inverse if $vt = 1$ for some $t \in R$,

- (a) Prove that u is a unit if and only if it has both a right and a left inverse.

Proof: Suppose $u \in R$ is a unit. By definition there exists some $r \in R$ such that $ru = ur = 1$. Therefore u has both a right and left inverse.

Suppose $u \in R$ has a right and left inverse. By definition there exists some $r, l \in R$ such that $ur = lu = 1$. Now consider r and note that $r = 1r = (lu)r = l(ur) = l1 = l$. Therefore u is a unit. □

- (b) Prove that if u has a right inverse then u is not a right zero divisor.

Proof: Suppose $u \in R$ such that there exists some $r \in R$ where $ur = 1$. For the sake of contradiction suppose u is a right zero divisor. Therefore there exists some $x \in R$ such that $xu = 0$ with $u, x \neq 0$. Note that $x = x(1) = x(ur) = (xu)r = 0r = 0$, a contradiction. □

- (c) Prove that if u has more than one right inverse then u is a left zero divisor.

Proof: Suppose $u \in R$ such that there exists $r_1, r_2 \in R$ where $r_1 \neq r_2$ and $ur_1 = ur_2 = 1$. Note that $r_1 - r_2 \neq 0$ since $r_1 \neq r_2$. Consider $u(r_1 - r_2) = ur_1 - ur_2 = 1 - 1 = 0$. Thus u is a left zero divisor. □

- (d) Prove that if R is a finite ring then every element that has a right inverse is unit.

Proof: Suppose R is finite and $x \in R$ such that there exist some $r \in R$ where $xr = 1$. We want to show that there exists some $l \in R$ such that $lx = 1$, then by (a), x is unit. Consider the map $\varphi : R \rightarrow R$ defined by $\varphi(l) = lx$. Choose $a, b \in R$ such that $\varphi(a) = \varphi(b)$. By definition we get that $ax = bx$, right multiplying by r we get that $a = b$ therefore φ is injective. Since R is finite φ must also be surjective. Then there exist some $l \in R$ such that $\varphi(l) = lx = 1$. □

Section 7.2

Exercise (7.2.2). Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be an element of the polynomial ring $R[x]$. Prove that $p(x)$ is a zero divisor in $R[x]$ if and only if there is a nonzero $b \in R$ such that $bp(x) = 0$.

Proof: Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be an element of the polynomial ring $R[x]$.

Clearly if there is a nonzero $b \in R$ such that $bp(x) = 0$ then $p(x)$ is a zero divisor in $R[x]$, since b is also in $R[x]$. Suppose $p(x)$ is a zero divisor in $R[x]$. Therefore there exists some $g(x) \in R[x]$ of minimal degree m with the form $g(x) = b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0$ such that $g(x) \neq 0$ and $p(x)g(x) = g(x)p(x) = 0$.

It then follows that

$$p(x)g(x) = \sum_{k=0}^{n+m} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k = 0.$$

Note that in the expanded sum the coefficient on x^{n+m} is $a_n b_m = 0$. Now, when we consider the polynomial $a_n(g(x))$ we get that the leading coefficient is $a_n b_m = 0$ so $a_n(g(x))$ is a polynomial of degree less than m . Now consider that,

$$p(x)(a_n g(x)) = \sum_{k=0}^{n+m} \left(\sum_{i=0}^k a_i (a_n b_{k-i}) \right) x^k = a_n \sum_{k=0}^{n+m} \left(\sum_{i=0}^k a_i (b_{k-i}) \right) x^k = a_n 0 = 0.$$

But since $g(x)$ was the minimal nonzero polynomial with the property that $p(x)g(x) = g(x)p(x) = 0$ it must be the case that $a_n g(x) = 0$

We will proceed by induction to show that $a_{n-j}g(x) = 0$ for all $0 \leq j < n$, the argument preceding this was the base case where $j = 0$. Suppose that some there exists some $0 \leq j < n$ where for all $r < j$ $a_{n-r}g(x) = 0$. Recall the sum

$$p(x)g(x) = \sum_{k=0}^{n+m} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k = 0$$

and consider the coefficient on $x^{m+(n-j)}$, we see that it is the following sum,

$$\left(\sum_{i=0}^{m+(n-j)} a_i b_{(m+(n-j))-i} \right) = 0$$

By the inductive hypothesis we know that for all products $a_i b_{(m+(n-j))-i}$ where $n-j < i \leq n$ we get $a_i b_{(m+(n-j))-i} = 0$. Clearly all products where $i > n$ will be zero since $p(x)$ is only order n . Consider $0 \leq i \leq n-j-1$ and note that these products will have a b term that is larger than b_m since $b_{m+(n-j)-(n-j-1)} = b_{m+1}$ so they also become zero since $g(x)$ is order m . Therefore we conclude that $a_{n-j} b_m = 0$ and thus $a_{n-j} g(x)$ is polynomial degree less than m with the property that $p(x)(a_{n-j} g(x)) = a_{n-j} p(x) g(x) = 0$. Since $g(x)$ is minimal we conclude that $a_{n-j} g(x) = 0$.

Thus by induction we have shown that $a_{n-j} g(x) = 0$ for all $0 \leq j \leq n$. Looking at the leading term it follows that $a_{n-j} b_m = 0$ for all $0 \leq j \leq n$ therefore $b_m p(x) = 0$ $\square$

Exercise (7.2.3). Define the $\det R[[x]]$ of formal power series in the indeterminate x with coefficients from R to be all formal infinite sums,

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

Define addition and multiplication of power series in the same way as for power series with real or complex coefficients.

a Prove that $R[[x]]$ is commutative ring with 1.

Proof: Suppose $p(x), q(x), g(x) \in R[[x]]$ such that

$$p(x) = \sum_{n=0}^{\infty} a_n x^n \text{ and } q(x) = \sum_{n=0}^{\infty} b_n x^n,$$

$$g(x) = \sum_{n=0}^{\infty} h_n x^n.$$

Note that in $(R[[x]], +)$, 0 is the identity element, and it's closed with respect to inverses since $(R, +)$ is a group and for every coefficients a_n in $p(x)$ we can construct a $-p(x)$ with coefficients $-a_n$. It is also associative since $(R, +)$ is a group we get $(p(x) + q(x)) + g(x)$ has coefficients $((a_n + b_n) + h_n) = (a_n + (b_n + h_n))$ which are the coefficient for $p(x) + (q(x) + g(x))$. Thus $(R[[x]], +)$ is a group. Note that $(R[[x]], +)$ is an abelian group since R is a

ring,

$$p(x) + q(x) = \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n + b_n) x^n = \sum_{n=0}^{\infty} b_n x^n + \sum_{n=0}^{\infty} a_n x^n = q(x) + p(x).$$

We also get that $(R, \times)$ is associative, since R is a ring we get that

$$(p(x) \times q(x)) \times g(x) = \sum_{n=0}^{\infty} (a_n b_n) (h_n) x^n = \sum_{n=0}^{\infty} (a_n) (b_n h_n) x^n = p(x) \times (q(x) \times g(x))$$

. We also get that distributive laws hold in $R[[x]]$, since R is a ring we get that

$$(p(x) + q(x)) \times g(x) = \sum_{n=0}^{\infty} (a_n + b_n) (h_n) x^n = \sum_{n=0}^{\infty} (a_n h_n + b_n h_n) x^n = (p(x) \times g(x) + q(x) \times g(x)).$$

So $R[[x]]$ is a ring. Since R is a commutative ring we get that,

$$(p(x) \times q(x)) = \sum_{n=0}^{\infty} \sum_{k=0}^n (a_k b_{n-k}) x^n = \sum_{n=0}^{\infty} \sum_{k=0}^n (b_k a_{n-k}) x^n = (q(x) \times p(x)).$$

Consider the polynomial in $R[[x]]$, such that $1(x) = 1 + 0x_1 + 0x_2 \dots$. Clearly $p(x) \times 1(x) = p(x)$. So $R[[x]]$ has 1. $\square$

b Show that $1 - x$ is a unit in $R[[x]]$ with inverse $1 + x_1 + x_2 + \dots$.

Proof: Expanding the product we get,

$$(1 - x)((1 + \sum_{i=1}^{\infty} x_i)) = (1 + \sum_{i=1}^{\infty} x_i) - x(1 + \sum_{i=1}^{\infty} x_i) = (1 + \sum_{i=1}^{\infty} x_i - \sum_{i=1}^{\infty} x_i) = 1.$$

$\square$

Exercise (7.2.4). Prove that if R is an integral domain then the ring of formal power series $R[[x]]$ is also an integral domain.

Proof: Suppose R is an integral domain, and let $p(x), q(x) \in R[[x]]$ such that $p(x), q(x) \neq 0$ and,

$$p(x) = \sum_{n=0}^{\infty} a_n x^n \text{ and } q(x) = \sum_{n=0}^{\infty} b_n x^n.$$

Now let $a_i x^i$ and $b_j x^j$ be the first pair of non zero terms in $p(x)$ and $q(x)$, i.e. x^i, x^j are minimal such that $a_i, b_j \neq 0$. Then the coefficient on the x^{i+j} term will be

$$\sum_{k=0}^{i+j} a_k b_{i+j-k} = a_i b_j$$

Since when $k > i$ $a_k = 0$ and when $k > i$ then $b_{i+j-k} = 0$. Recall that since R is an integral domain $a_i b_j \neq 0$. Therefore the product $p(x)q(x) \neq 0$ where $p(x), q(x) \neq 0$, thus $R[[x]]$ has no zero divisors. We have showed that $R[[x]]$ is a commutative ring with identity, therefore $R[[x]]$ is an integral domain. $\square$

Exercise (7.2.9ac). let $\alpha = r + r^2 - 2s$ and $\beta = -3r^2 + rs$ be the two elements of the integral group ring $\mathbb{Z}D_8$ described in this section. Compute the following elements of $\mathbb{Z}D_8$,

(a) $\beta\alpha$

Proof:

$$\begin{aligned} \beta\alpha &= (-3r^2 + rs)(r + r^2 - 2s) \\ &= -3r^2(r + r^2 - 2s) + rs(r + r^2 - 2s) \\ &= -3r^2r - 3r^2r^2 + (-3r^2)(-2s) + rsr + rsr^2 - rs2s \\ &= -3r^3 - 3r^4 + 3r^2(2s) + rsr + rsr^2 - rs(2s) \\ &= -3r^3 - 3(1) + 6r^2s + (sr^{-1})r + (sr^{-1})r^2 - 2rs^2 \\ &= -3r^3 - 3 + 6r^2s + s + sr - 2r \\ &= -3r^3 + 6r^2s - 2r + sr + s - 3 \end{aligned}$$

$\square$

(c) α^2

Proof:

$$\begin{aligned}
 \alpha^2 &= (r + r^2 - 2s)(r + r^2 - 2s) \\
 &= r(r + r^2 - 2s) + r^2(r + r^2 - 2s) - 2s(r + r^2 - 2s) \\
 &= rr + rr^2 - r2s + r^2r + r^2r^2 - r^22s - 2sr - 2sr^2 + 2s2s \\
 &= r^4 + 2r^3 - 2r^3s + r^2 - 4r^2s - 2rs + 4
 \end{aligned}$$

□

Exercise (7.3.1). Prove that rings $2\mathbb{Z}$ and $3\mathbb{Z}$ are not isomorphic.

Proof: For the sake of contradiction suppose that there exists an isomorphism $\varphi : 2\mathbb{Z} \rightarrow 3\mathbb{Z}$.

Therefore there exists some $r \in 3\mathbb{Z}$ such that $\varphi(2) = r$. Note that $\varphi(4) = \varphi(2)\varphi(2) = r^2$ and $\varphi(4) = \varphi(2 + 2) = \varphi(2) + \varphi(2) = 2r$. Therefore we get that $r^2 = 2r$ and since $r = 3x$ for some $x \in \mathbb{Z}$ we get $9x^2 = 6x$ and since $\mathbb{Z}$ has no zero divisors we have cancellation so $9x = 6$ which has no integer solution. Thus $2\mathbb{Z}$ and $3\mathbb{Z}$ are not isomorphic. □

Exercise (7.3.4). Find all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}/30\mathbb{Z}$. In each case describe the kernel and the image.

Proof: To begin a ring homomorphism, must be a group homomorphism on the additive groups. So as we have seen every group homomorphism is uniquely determined by the mapping of $1 \in \mathbb{Z}$. (Let σ and β be homomorphisms such that $\sigma(1) = \beta(1)$. Then $\sigma(n) = \sigma(1^n) = \sigma(1)^n = \beta(1)^n = \beta(n)$). Therefore the set of all ring homomorphisms is contained in the set of all additive group homomorphism. Let σ_g be the group homomorphism which maps $\sigma_g(1) = g$. It must also be the case that for a ring homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/30\mathbb{Z}$, we get $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$. Now note that $\varphi_g(ab) = \varphi_g(\sum_1^{ab} 1) = \sum_1^{ab} \varphi_g(1) = \sum_1^{ab} g = (ab)g$ and $\varphi_g(a)\varphi_g(b) = \varphi_g(\sum_1^a 1)\varphi_g(\sum_1^b 1) = (\sum_1^a \varphi_g(1))(\sum_1^b \varphi_g(1)) = (\sum_1^a g)(\sum_1^b g) = (ag)(bg) = (ab)g^2$

Therefore in order for φ_g to be a ring homomorphism $(ab)g = (ab)g^2$ for all $ab \in \mathbb{Z}$, which follows if $g = g^2$ in $\mathbb{Z}/30$.

So elements $x \in \mathbb{Z}/30\mathbb{Z}$ which satisfy $x = x^2$ are $\bar{1}, \bar{6}, \bar{10}, \bar{15}, \bar{21}, \bar{25}$. $\square$